

Master's Thesis

Leakage suppression via pulse
waveform optimization in
superconducting quantum systems

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July 27th, 2026

内容梗概

超伝導トランズモン量子ビットは高速かつ高忠実度な制御パルスが必要とするが、それは必然的に非計算状態へのスペクトル漏れ（リーク）を誘発し、ゲート性能を低下させる。本論文では、これらの漏れ誤差を抑制し、単一および2量子ビットの量子演算を高速化するための高度な波形最適化手法を提案する。単一量子ビットのXゲートにおいては、再帰的なDRAG-1およびDRAG-2条件を、フーリエ変換がゼロになる要件として数学的に再定式化した。この再定式化により、エコー法を用いることで、従来のシード関数を介さずに新規の解析的なフラットトップパルスを直接合成し、短いゲート時間における漏れ誤差の抑制に成功した。2量子ビットのCZゲートにおいては、漏れが少なくゲート時間が比較的長い演算向けに、従来のSlepian（スレピアン）パルスの代替として反対称チェビシェフパルスを提案する。さらに、元の問題を再定式化することに基づく最適化を活用し、標的とするスペクトル抑制を達成した。QuTiPを用いた数値シミュレーションにより、提案するDRAG-1パルスが高い忠実度を維持しつつ、理論上の最小ゲート時間を約20%短縮することが確認された。

Leakage suppression via pulse waveform optimization in superconducting quantum systems

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Abstract

Superconducting transmon qubits require fast, high-fidelity control pulses that inevitably induce spectral leakage into non-computational states, degrading gate performance. This thesis introduces advanced waveform optimization techniques to suppress these leakage errors and accelerate single- and two-qubit quantum operations. For the single-qubit X gate, the recursive DRAG-1 and DRAG-2 conditions are mathematically reformulated as zero Fourier transform requirements. Using the echo method, a novel analytical flat-top pulse is synthesized directly, bypassing conventional seed functions, successfully suppressing leakage error for short gate time. For the two-qubit CZ gate, an anti-symmetric Chebyshev pulse is proposed as an alternative to traditional Slepian pulses for a low leakage longer gate time operations. Furthermore, optimization based on reformulating the original problem is utilized to achieve targeted spectral suppression. Numerical simulations via QuTiP confirm that the proposed DRAG-1 pulse reduces theoretical minimum gate times by approximately 20% while maintaining high fidelity.

Table 1: Notation List

Symbol	Description
α, β	Complex probability amplitudes of the basis states
$H(t)$	Total time-dependent system Hamiltonian
$\mathcal{T}$	Dyson time-ordering operator
$\mathcal{F}$	Fourier transform
T	Total gate duration
τ_d	gate duration in non-linear time frame.
t_d	delay time in the echo method (X gate) gate duration in the lab frame (CZ)
g	Capacitive coupling strength between the two qubits (in X gate)
g	Pulse designed in non-linear time frame (CZ)
$\hat{a}^\dagger, \hat{a}$	Creation and annihilation operators
$\lambda_{k,m}$	Transition matrix elements, defined as $\langle k \hat{n} m\rangle$
λ_k	$\lambda_{k-1,k}$ (X gate)
$\lambda_k(N, W)$	Slepian eigenvalue representing the fractional energy concentration
ω_d	Microwave drive frequency
$\omega_{i,j}$	Transition frequency from the i -th state to the j -th ($E_j - E_i$)
ω_k	$\omega_{0,k}$ Transition frequency of the k -th state ($E_k - E_0$) (X gate)
ω_i	Tunable transition frequency of Qubit i (CZ)
α_i	Anharmonicity of Qubit i (in CZ gate framework)
$\tilde{\Delta}_k$	Detuning of the k -th level, defined as $\omega_k - k\omega_d$
Δ_k	$\omega_k - k\omega_1$
Δ	Often used as shorthand for the anharmonicity Δ_2
$\delta(t)$	² Time-dependent phase ramping (detuning) to compensate for AC Stark shifts

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Contents

Chapter 1	Introduction	1
1.1	Background and Motivation	1
1.2	Problem Statement	1
1.3	Objectives and Contributions	2
1.4	Thesis Organization	3
Chapter 2	Theoretical Foundations: Single-Qubit X Gate	4
2.1	Introduction to the Quantum Bit	4
2.2	Gate Fidelity and Leakage	5
2.3	Qudit Hamiltonian in the Rotating Frame	7
2.4	Approximation of λ	9
2.5	Schrieffer-Wolff Transformation	9
2.6	Derivative Removal by Adiabatic Gate (DRAG)	10
2.6.1	The DRAG-1 Scheme	10
2.6.2	The DRAG-2 Scheme	11
2.7	Non-perturbative Methods	11
2.7.1	NPAD	11
2.8	Block Diagonalization via Riccati Engineering	12
Chapter 3	Theoretical Foundations: Two-Qubit CZ gate	14
3.1	SQUID loop and Z -control	14
3.2	Qubit Architecture	15
3.3	Adiabatic Implementation	15
3.3.1	Calculation of Leakage Error	16
3.3.2	Dynamical and Conditional Phase Accumulation	17
3.3.3	Nonlinear time transformation	17
3.4	Slepian pulses	18
3.5	Limitations of Slepian Pulses	18
Chapter 4	Proposed Waveform Design Techniques	19

4.1	Inverse Design of the DRAG-1 Pulse	19
4.2	Inverse Design of the DRAG-2 Pulse	20
4.3	Pulse Design	22
4.3.1	DRAG-1 pulse	22
4.3.2	DRAG-2 pulse	23
4.4	Optimization via Riccati Engineering	23
4.4.1	Variational Pulse Adjustment	24
4.5	CZ gate	25
4.5.1	The Anti-symmetric Chebyshev Pulse	26
4.5.2	Spectrum Suppression via Sequential Quadratic Pro- gramming	26
4.5.3	Redefining the Slepian eigenvalue problem	27
4.5.4	Sidelobe-Aware Optimization	28
Chapter 5	Simulation and Numerical Results	29
5.1	X gate	29
5.1.1	Fidelity	29
5.1.2	Pulse shapes	30
5.1.3	Performance	31
5.1.4	Minimum Gate Time of Our DRAG-1	31
5.1.5	Limitations on our proposed design of DRAG-2 Pulse from the Echo Method	32
5.1.6	Non-perturbative Validation of the Y-DRAG Correction	33
5.1.7	Hardware Constraint	33
5.2	Riccati Engineering	34
5.3	CZ gate	35
5.3.1	Simulation	35
5.3.2	Anti-symmetric Chebyshev pulse	35
5.3.3	Sidelobe-Aware Optimization	35
5.3.4	Slepian pulse with redefined maximization problem . . .	35
5.3.5	Performance	37
5.3.6	Minimum Gate time	38

Chapter 6	Conclusion	39
6.1	Future Work	40
	Acknowledgments	41
	References	42
	Appendix	A-1
A.1	SWTs to Derive DRAG-2 Formula	A-2
A.2	Wolfram Code for Validating Y DRAG Correction without Per- turbation	A-3

Chapter 1 Introduction

1.1 Background and Motivation

Quantum computing represents a paradigm shift in computational science, offering exponential speedups for specific classes of problems, including quantum chemistry simulation [15], cryptography [2], and large-scale optimization [33] in many industry and scientific areas. At the heart of this physical realization are quantum bits (qubits). Among the various physical implementations, superconducting qubits—particularly the transmon architecture—have established themselves as a leading foundational platform [10]. Their macroscopic nature allows for strong coupling to microwave control fields and flexible lithographic fabrication, making them highly scalable [3]. However, realizing fault-tolerant quantum computation requires extremely high-fidelity quantum operations. Current Noisy Intermediate-Scale Quantum (NISQ) devices are fundamentally limited by decoherence and operational errors. For superconducting qubits, single- and two-qubit gate fidelities are highly sensitive to the precise shape of the microwave control pulses used to drive the transitions. Imperfect control pulses lead to phase errors, over-rotation, and leakage out of the computational subspace, severely degrading the overall performance of the quantum processor.

1.2 Problem Statement

The transmon qubit is inherently a weakly anharmonic oscillator. Because the energy difference between the $|0\rangle \leftrightarrow |1\rangle$ transition and the $|1\rangle \leftrightarrow |2\rangle$ transition is relatively small, applying fast, high-amplitude microwave pulses to minimize decoherence time inevitably results in spectral leakage into higher, non-computational energy states. The Derivative Removal by Adiabatic Gate (DRAG) technique is a highly successful methodology for suppressing this leakage. However, the demand for even higher fidelities necessitates further refinement of the underlying waveforms used within the DRAG framework. Because standard Gaussian or square base pulses can be spectrally broad or susceptible to noise, there is a critical need to integrate mathematically optimal pulse

shapes into the DRAG protocol. Specifically, by actively suppressing the amplitude of these optimized waveforms, this combined approach can strictly confine the energy spectrum while maintaining fast gate times. Implementing this requires robust calibration frameworks to validate these amplitude-suppressed pulses against realistic Hamiltonian dynamics.

1.3 Objectives and Contributions

The primary objective of this thesis is to investigate, optimize, and evaluate advanced gate pulse waveforms for superconducting transmon qubits, both one qubit and two qubit operations. By uncovering the underlying spectral properties of established control techniques and introducing novel waveform designs, this work minimizes leakage errors while accelerating gate execution times. The specific original contributions of this thesis are detailed as follows:

Single-Qubit Gate Innovations

Spectral Formulation of Recursive DRAG Successfully demonstrated that the condition equations for DRAG-1 and DRAG-2 pulses can be rigorously defined as a zero Fourier transform requirement. This connects the superconducting DRAG formalism to analogous mathematical conditions frequently observed in spin qubit research [5, 12, 30, 31], providing a deeper physical intuition for recursive DRAG schemes.

Novel Pulse Design Methodology Leveraged this newly established Fourier transform formulation to develop a novel analytical pulse design approach. This method bypasses conventional constraints and provides an improved control over the resulting pulse shape, successfully achieving shorter single-qubit gate durations without compromising operational fidelity.

Two-Qubit Gate Innovations

Anti-Symmetric Chebyshev Pulses Discovered and proposed a novel control waveform, designated as the anti-symmetric Chebyshev pulse, tailored to optimize two-qubit gate operations.

Targeted Spectral Suppression Adapted and extended the traditional Slepian pulse optimization framework by applying a technique specifically designed to minimize the Fourier transform amplitude within a precisely targeted

frequency range, thereby optimizing the waveform for two-qubit leakage suppression.

1.4 Thesis Organization

The remainder of this thesis is systematically structured to guide the reader from the fundamental physical principles of superconducting quantum systems to the mathematical modeling and numerical validation of the proposed waveform optimization techniques. Chapter 2 establishes the theoretical foundations for the single-qubit X gate, providing an overview of qubit dynamics, leakage errors, and conventional mitigation frameworks, including the Schrieffer-Wolff transformation, recursive DRAG schemes, and Riccati engineering. Chapter 3 extends this theoretical groundwork to two-qubit operations, detailing the architecture, adiabatic implementation, and phase accumulation of the CZ gate, while analyzing the characteristics and limitations of traditional Slepian pulses. Building upon these foundations, Chapter 4 introduces the novel waveform design techniques proposed in this work. For single-qubit control, it details a new analytical pulse design methodology using the echo method and variational adjustments via Riccati engineering. For two-qubit operations, it presents the formulation of the anti-symmetric Chebyshev pulse and introduces targeted spectral suppression techniques utilizing Sequential Quadratic Programming. Chapter 5 presents comprehensive simulation data and numerical results, evaluating the fidelity and physical viability of the proposed pulses against hardware constraints. It includes a non-perturbative validation of the Y -DRAG correction and compares the performance of the newly designed waveforms against conventional standard pulses for both X and CZ gates. Finally, Chapter 6 summarizes the primary findings and contributions of the thesis, concluding with an outline of prospective avenues for future research.

Chapter 2 Theoretical Foundations: Single-Qubit X Gate

In this chapter, we establish the theoretical foundations and review legacy methodologies critical to understanding the fast control pulse design method (X gate) proposed in this work. We first provide an overview of transmon qubits to fix our notation and convention. Subsequently, we examine the DRAG formalism and its recursive variants, Schrieffer-Wolff transformations, and Riccati engineering.

2.1 Introduction to the Quantum Bit

The transition from classical to quantum information processing represents a fundamental paradigm shift in computational theory. In classical computing, the foundational unit of information is bit which exists in a mutually exclusive, deterministic state of either 0 or 1. Quantum computing replaces this discrete entity with the quantum bit, or qubit, an entity governed by the principles of quantum mechanics. Rather than being confined to a single state, a qubit can exist in a coherent superposition of both states simultaneously, unlocking exponentially larger state spaces and enabling computational speedups for specific classes of problems, such as integer factorization [2] and molecular simulation [15].

Mathematically, the state of a single, isolated qubit is represented as a state vector in a two-dimensional complex Hilbert space. Using Dirac notation, the computational basis states are denoted as $|0\rangle$ and $|1\rangle$, corresponding to the classical 0 and 1. A pure qubit state $|\psi\rangle$ is expressed as a linear combination of these basis states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

Here, α and β are complex probability amplitudes satisfying $|\alpha|^2 + |\beta|^2 = 1$. This constraint allows the pure state of a qubit to be mapped onto the surface of a unit sphere in three-dimensional space, known as the Bloch sphere. On the Bloch sphere, the poles represent the basis states $|0\rangle$ and $|1\rangle$, while points on the equator represent equal superpositions with varying relative phases. This

geometric representation is indispensable for visualizing single-qubit quantum gates as rotations about the sphere’s axes.

As a quantum mechanical system, a qubit’s dynamics are governed by the time-dependent Schrödinger equation. Consequently, any control applied to the system is mathematically represented by the Hamiltonian operator driving this evolution. In formula, [29]

$$\psi(t) = \mathcal{T} \left(\exp \left[\int_0^T iH(t) dt \right] \right) \psi(0).$$

Here, $\mathcal{T}$ denotes the Dyson time-ordering operator, T is the total gate duration, and $H(t)$ is the system’s total Hamiltonian. In practice, $H(t)$ consists of a static drift term defining the unperturbed system and a time-dependent control term corresponding to external driving fields, such as shaped microwave pulses. The primary objective of quantum control is to engineer these time-dependent waveforms so that the resulting evolution operator accurately synthesizes a desired target unitary gate on the computational subspace.

2.2 Gate Fidelity and Leakage

Physical transmons are weakly anharmonic oscillators, meaning their $\{|0\rangle, |1\rangle\}$ computational subspace is not entirely isolated from higher energy levels, most notably the $|2\rangle$ state. While fast quantum operations require short, high-bandwidth microwave pulses, their broad frequency spectra can inadvertently drive transitions into these non-computational states. As illustrated in Fig. 1, leakage creates a fundamentally different error profile compared to coherent calibration errors. By driving population into higher energy states, leakage causes non-unitary evolution within the computational subspace. This manifests as a mixed state ρ , a severe error that degrades gate fidelity and crucially evades standard quantum error correction protocols.

Therefore, achieving the high-fidelity gates required for fault tolerance demands advanced waveform optimization. The control fields defined in $H(t)$ must be precisely shaped to balance two competing physical constraints: executing operations as rapidly as possible to minimize decoherence, while strictly suppressing the spectral components that induce leakage.

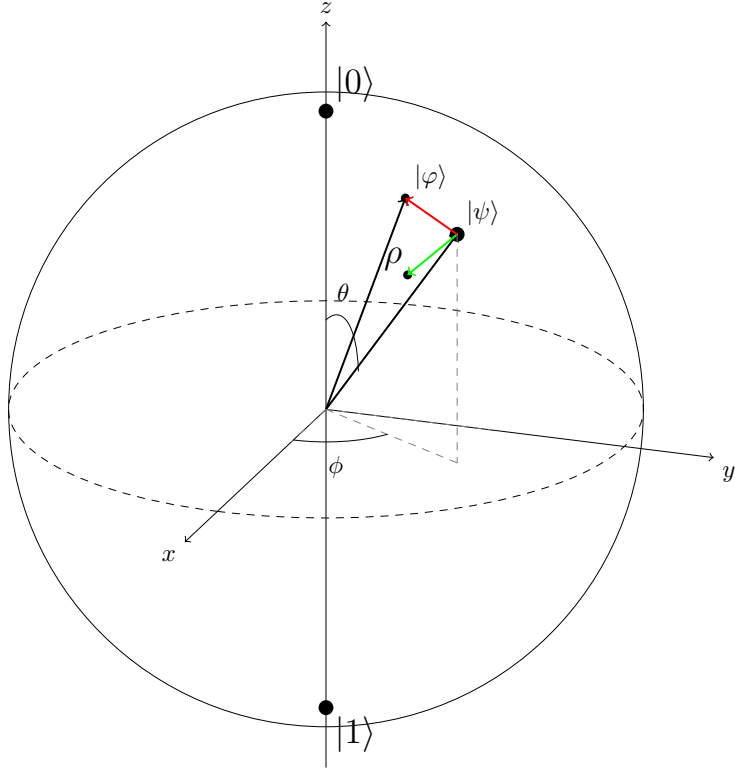


Figure 1: Geometric representation of quantum gate errors on the Bloch sphere. An ideal gate operation targets the pure state $|\psi\rangle$. Coherent errors result in an unintended pure state $|\varphi\rangle$ on the surface of the sphere, where the divergence between $|\varphi\rangle$ and $|\psi\rangle$ contributes directly to gate infidelity. Conversely, population leakage into non-computational higher energy states results in a mixed state ρ situated within the interior of the sphere.

2.3 Qudit Hamiltonian in the Rotating Frame

Because a transmon is inherently an anharmonic oscillator, it possesses higher excited states beyond $|0\rangle$ and $|1\rangle$. Consequently, it acts as a qudit—a quantum system with d levels—rather than an ideal qubit, which inevitably introduces leakage into unwanted noncomputational states.

To model this leakage analytically, we review the derivation of the qudit Hamiltonian truncated at the first $N_{\max} + 1$ levels of the transmon by following Ref. [14]. In passing, we derive the Hamiltonian without the Rotating Wave Approximation (RWA)[6].

The Hamiltonian of a transmon qudit in the lab frame is [32]

$$\hat{H}_{\text{lab}} = 4E_c(\hat{n} - \hat{n}_g)^2 - E_J \cos(\hat{\varphi}), \quad (1)$$

where $\hat{n}_g$ denotes the offset charge, $\hat{n}$ the number operator, E_C the charging energy, E_J the Josephson energy, and φ the phase operator. From the (non-degenerate) perturbation theory, we find that the eigenstates of this Hamiltonian are harmonics with non-uniform energy spacings. We denote the k -th excited state as $|k\rangle$ and its φ -representation $\psi_k(\varphi)$. Now we consider the expansion of $\hat{n}$ in terms of $|k\rangle\langle m|$ ($k \geq 0, m \geq 0$). Note that k 's parity coincides with ψ_k 's. Since $\hat{n}$ flips the sign, the coefficient

$$\lambda_{k,m} = \langle k | \hat{n} | m \rangle = \int_{-\pi}^{\pi} \psi_k^*(\varphi) \left(-i \frac{\partial}{\partial \varphi} \right) \psi_m(\varphi) d\varphi \quad (2)$$

is zero unless k and m are of opposite parity. Hence $\hat{n}$ can be expressed as

$$\begin{aligned} \hat{n} = & \sum_{k=0}^{N_{\max}-1} \left[\lambda_{k,k+1} |k\rangle\langle k+1| \right. \\ & \left. + \sum_{j=1} \lambda_{k,k+2j+1} |k\rangle\langle k+2j+1| \right] + \text{h.c.} \end{aligned} \quad (3)$$

If we apply the change-of-basis V on the Hilbert space, the Hamiltonian changes as

$$H' = V H V^\dagger + i \dot{V} V^\dagger, \quad (4)$$

where the last term is called a gauge term. Applying the diagonal frame transformation with

$$R^\dagger = \exp \left(i \omega_d t \sum_{k=0}^{N_{\max}} k |k\rangle\langle k| \right) \quad (5)$$

yields the rotating frame Hamiltonian

$$\begin{aligned} \hat{H}_{\text{rot}} = & \sum_{k=0}^{N_{\text{max}}} \Delta_k |k\rangle \langle k| + \Omega_{\text{RF}}(t) \left[\right. \\ & \sum_{k=0}^{N_{\text{max}}-1} \sum_{j=1}^{N_{\text{max}}-1} \lambda_{k,k+2j+1} |k\rangle \langle k+2j+1| e^{-(2j+1)i\omega_d t} \\ & \left. + \sum_{k=0}^{N_{\text{max}}-1} \lambda_{k,k+1} |k\rangle \langle k+1| e^{-i\omega_d t} + \text{h.c.} \right], \end{aligned} \quad (6)$$

where $\Delta_k = \omega_k - k\omega_d$, ω_d is the drive frequency and $\omega_k = E_k - E_0$ (eigenenergy differences between the ground state and k -th excited state). Henceforth, when we are discussing one-qubit gates, if we simply write Δ this means Δ_2 . Note that Δ is the anharmonicity of the qubit.

If we apply RWA to every rotating term, Eq. (6) reduces to

$$\begin{aligned} \hat{H}_{\text{RWA}} = & \sum_{j=1}^{N_{\text{max}}} (\Delta_j + j\delta(t)) |j\rangle \langle j| \\ & + \sum_{j=1}^{N_{\text{max}}} \lambda_j \left[\frac{\Omega_x(t)}{2} \hat{\sigma}_{j-1,j}^x + \frac{\Omega_y(t)}{2} \hat{\sigma}_{j-1,j}^y \right], \end{aligned} \quad (7)$$

where $\lambda_j = \lambda_{j-1,j}$, $\hat{\sigma}_{j-1,j}^x = |j-1\rangle \langle j| + |j\rangle \langle j-1|$, and $\hat{\sigma}_{j-1,j}^y = -i|j-1\rangle \langle j| + i|j\rangle \langle j-1|$.

The matricial form when $N_{\text{max}} = 3$ is

$$\begin{pmatrix} 0 & \frac{1}{2}\lambda_1(\Omega_x(t) - i\Omega_y(t)) & 0 & 0 \\ \frac{1}{2}\lambda_1(\Omega_x(t) + i\Omega_y(t)) & \delta(t) + \Delta_1 & \frac{1}{2}\lambda_2(\Omega_x(t) - i\Omega_y(t)) & 0 \\ 0 & \frac{1}{2}\lambda_2(\Omega_x(t) + i\Omega_y(t)) & 2\delta(t) + \Delta_2 & \frac{1}{2}\lambda_3(\Omega_x(t) - i\Omega_y(t)) \\ 0 & 0 & \frac{1}{2}\lambda_3(\Omega_x(t) + i\Omega_y(t)) & 3\delta(t) + \Delta_3 \end{pmatrix}. \quad (8)$$

Including the first order corrections of RWA, the Hamiltonian becomes

$$\begin{aligned} \hat{H}_{\text{RWA},1} = & \sum_{k=0}^{N_{\text{max}}} \tilde{\Delta}_k |k\rangle \langle k| + \frac{1}{2} (\Omega_x - i\Omega_y) \left[\right. \\ & \sum_{k=0}^{N_{\text{max}}-1} \lambda_{k,k+3} |k\rangle \langle k+3| e^{-2i\omega_d t} \\ & \left. + \sum_{k=0}^{N_{\text{max}}-1} \lambda_{k,k+1} |k\rangle \langle k+1| \right] + \text{h.c.} \end{aligned} \quad (9)$$

where, for brevity, the last h.c. only refers off-diagonal components.

2.4 Approximation of λ

In the quantum harmonic oscillator (QHO) approximation, $\hat{n}$ is expressed in terms of the annihilation and creation operators:

$$\hat{n} \approx in_{\text{zpf}}(\hat{a}^\dagger - \hat{a}).$$

Recall $\hat{a} = \sum_{k=0}^{\infty} \sqrt{k+1} |k\rangle \langle k+1|$,

$$\langle k+1 | \hat{a}^\dagger | k \rangle = \sqrt{k+1} \quad \text{and} \quad \langle k+1 | \hat{a} | k \rangle = 0.$$

Hence,

$$\lambda_{k,k+1} \approx \langle k+1 | \hat{n} | k \rangle = in_{\text{zpf}} \sqrt{k+1}.$$

The prefactor in_{zpf} could be regarded as absorbed by the drive amplitude $\Omega_{\text{RF}}(t)$.

Hence we simply set

$$\lambda_{k,k+1} \approx \sqrt{k+1}.$$

For λ_1 is entirely absorbed into $\Omega_{\text{RF}}(t)$, it is exactly (not approximately) $\lambda_1 = 1$.

Likewise $\lambda_{k,k+3}$ can be computed as [17]

$$\lambda_{k,k+3} \approx c \sqrt{(k+1)(k+2)(k+3)},$$

with a proportionality constant c .

2.5 Schrieffer-Wolff Transformation

While the rotating frame transformation in Eq. (6) is computed exactly, such exact analytical transformations are not always achievable. In general, the generator S is chosen so that $i\dot{S}$ cancels the error term in the Hamiltonian. In Eq. (4), V is assumed to be a unitary operator. we now consider a scenario where the transformation operator V is parameterized by an anti-Hermitian operator S . Specifically, V is defined exponentially as:

$$V = \exp(S),$$

we can expand the Hamiltonian in a power series of S as

$$H' = i\dot{S} + H + [S, H] + \frac{1}{2}[S, [S, H]] + \dots \quad (\text{Hadamard's lemma}) \quad (10)$$

This transformation is what is called Schrieffer-Wolff transformation (SWT). It is important to emphasize that the expansion is mathematically rigorous and holds generally true for any generator S and Hamiltonian H , provided the system is defined within a finite-dimensional Hilbert space. While truncating the series early causes inaccuracies, adding more terms makes subsequent computations excessively complex. In [Chapter 4](#), we develop a method to elongate the range of gate duration for which the approximation is valid.

2.6 Derivative Removal by Adiabatic Gate (DRAG)

The Derivative Removal by Adiabatic Gate (DRAG) technique is a highly effective method for suppressing leakage. It works by applying a secondary corrective pulse—proportional to the time derivative of the primary pulse envelope—along an orthogonal control axis, which actively cancels unwanted transitions to higher energy levels (such as the $|2\rangle$ state in transmon qubits) and corrects phase errors. Gambetta and Motzoi [\[7\]](#) derived the DRAG Y correction formula

$$\Omega_y = -\frac{1}{\Delta}\dot{\Omega}_x, \quad (11)$$

and demonstrated that that AC Stark shift can be compensated by way of phase ramping,

$$\delta(t) = \frac{4 - \lambda^2}{4\Delta} \Omega_x^2. \quad (12)$$

Collectively [Eq. \(11\)](#) and [Eq. \(12\)](#) are referred to as DRAG conditions. When these two conditions are applied within the weak driving regime, the Hamiltonian becomes

$$\hat{\mathcal{H}}_1(t) \approx \frac{\Omega_x}{2} \hat{\sigma}_{0,1}^x + \frac{\lambda_2 \Omega_x^2}{8\Delta_2} \hat{\sigma}_{0,2}^x - \Omega_x^2 \left[\frac{(2\Delta_2 - \Delta_3)}{8\Delta_2^2} \right] \lambda_2 \lambda_3 \hat{\sigma}_{1,3}^x, \quad (13)$$

subject to the following boundary conditions:

$$\Omega_x(t) = \dot{\Omega}_x(t) = 0 \quad \text{at } t = 0, T.$$

2.6.1 The DRAG-1 Scheme

The Hamiltonian in [Eq. \(13\)](#) contains additional two-photon transition terms between the $|0\rangle \leftrightarrow |2\rangle$ and $|1\rangle \leftrightarrow |3\rangle$ states. SWTs can be applied to further

reduce these transitions. We now introduce a recursive DRAG method known as DRAG-1 [19] or R1D [14].

Applying Schrieffer-Wolff transformations with $S_2 = -i\lambda_2\Omega_1^2/(8\Delta_2^2)\hat{\sigma}_{0,2}^y$ and $S_3 = i\lambda_2\dot{\Omega}_1\Omega_1/(4\Delta_2^3)\hat{\sigma}_{0,2}^x$ to Eq. (13) yields

$$\hat{\mathcal{H}}_{\text{new}}(t) = \frac{\Omega_x}{2}\hat{\sigma}_{0,1}^x + \frac{\lambda_2\Omega_x^2}{8\Delta_2}\hat{\sigma}_{0,2}^x - \Omega_x^2 \left[\frac{(2\Delta_2 - \Delta_3)}{8\Delta_2^2} \right] \lambda_2\lambda_3\hat{\sigma}_{1,3}^x - \lambda_2 \sigma_{0,2}^x.$$

Therefore, by putting

$$\Omega_x(t) = \sqrt{\Omega_1^2 + \frac{2}{\Delta_2^2}(\dot{\Omega}_1^2 + \ddot{\Omega}_1\Omega_1)}, \quad (14)$$

we can cancel out the $|0\rangle \leftrightarrow |2\rangle$ leakage term. Here, Ω_1 must satisfy the boundary conditions $\Omega_1 = 0$ at $t = 0, T$.

For the X gate, the target condition

$$\int_0^T \Omega_x(t) dt = \pi \quad (15)$$

gets a slight modification if we consider terms up to the third degree in the frame transformation of Eq. (44)

$$\Omega'_x(t) = \Omega_x(t) - \frac{(4 - \lambda_2^2)\Omega_x^3(t)}{8\Delta^2}.$$

Corrections of this sort in the DRAG scheme are called superlinear corrections. We apply this correction in our simulations.

2.6.2 The DRAG-2 Scheme

The DRAG-2 scheme further removes the errors from the transformed Hamiltonian obtained in the previous subsection. In the same spirit, applying the appropriate SWTs (See A.1) removes $|1\rangle \leftrightarrow |3\rangle$ transition and

$$\Omega_1(t) = \sqrt{\Omega_2^2 + \frac{2}{\Delta_2^2}(\dot{\Omega}_2^2 + \ddot{\Omega}_2\Omega_2)}. \quad (16)$$

with boundary conditions $\Omega_2 = 0$ at $t = 0, T$.

2.7 Non-perturbative Methods

2.7.1 NPAD

A lot of efforts for leakage reduction and fidelity improvements have been done in the perturbative regime (i.e. $\Omega/\Delta \ll 1$). One of the framework advocated

that are also applicable in the non-perturbative strong driving regime is NPAD [20]. NPAD is an iterative method to diagonalize a nonperturbative Hamiltonian by applying static Givens rotations until the desired accuracy is achieved. While this method is only effective if the Hamiltonian has a rather simple form and only applicable to time-independent Hamiltonian, theoretically this is a generalization of recursive SWT in the perturbative regime.

2.8 Block Diagonalization via Riccati Engineering

Although Riccati engineering is a well-established mathematical tool in quantum control, it has not yet been extensively applied to the time-dependent control of single-qubit gates, with prior investigations primarily restricted to time-independent scenarios[8, 9]. Given a Hamiltonian partitioned as

$$H(t) = \begin{pmatrix} H_{11}(t) & H_{12}(t) \\ H_{21}(t) & H_{22}(t) \end{pmatrix},$$

with $H_{11}(t)$ 2×2 matrix and $H_{22}(t)$ 1×1 , we seek block diagonalizing transformation $U(t)$ according to Eq. (4) that distinctly separates the computational subspace from the non-computational.

The physical motivation for this approach is straightforward: if $H(t)$ can be exactly block-diagonalized, the dynamical evolution of a system initialized in the computational subspace (such as $|0\rangle$) will remain perfectly confined within that subspace in the new frame. Consequently, leakage to higher energy levels is strictly suppressed as long as the off-diagonal coupling terms are dynamically maintained at zero. As demonstrated by Gardas [9], the specific unitary transformation $U(t)$ that achieves this decoupling is

$$U(t) = \begin{pmatrix} (I_2 + X^\dagger X)^{-1/2} & X^\dagger(I_{n-2} + XX^\dagger)^{-1/2} \\ -X(I_2 + X^\dagger X)^{-1/2} & (I_{n-2} + XX^\dagger)^{-1/2} \end{pmatrix},$$

with $X(t) = (x_1(t) \ x_2(t))$ satisfying the forward matrix Riccati differential equation

$$i\dot{X} = H_{21} + H_{22}X - XH_{11} - XH_{12}X. \quad (17)$$

Under this frame change, the effective Hamiltonian governing the decoupled

computational subspace, $H'_{11}(t)$, takes the form

$$H'_{11}(t) = \left(I_2 + X^\dagger X\right)^{1/2} (H_{11} + H_{12}X)(I_2 + X^\dagger X)^{-1/2} - (I_2 + X^\dagger X)^{1/2} \partial_t \left[(I_2 + X^\dagger X)^{-1/2}\right]. \quad (18)$$

The complexity of the above formula is obviously one of the factors defying an application to time-dependent cases. In [4.4](#), we seek a method to bypass this complexity.

Chapter 3 Theoretical Foundations: Two-Qubit CZ gate

This section establishes the theoretical framework and physical architecture necessary for understanding and implementing the two-qubit CZ gate. We examine the operational principles of the adiabatic CZ gate and discuss why the Slepian pulse shape has traditionally been favored for control pulses. We use some symbols for different purposes in the X and CZ gates. When confusion arises, consult the notation table in Table. ?? located immediately following the abstract.

3.1 SQUID loop and Z -control

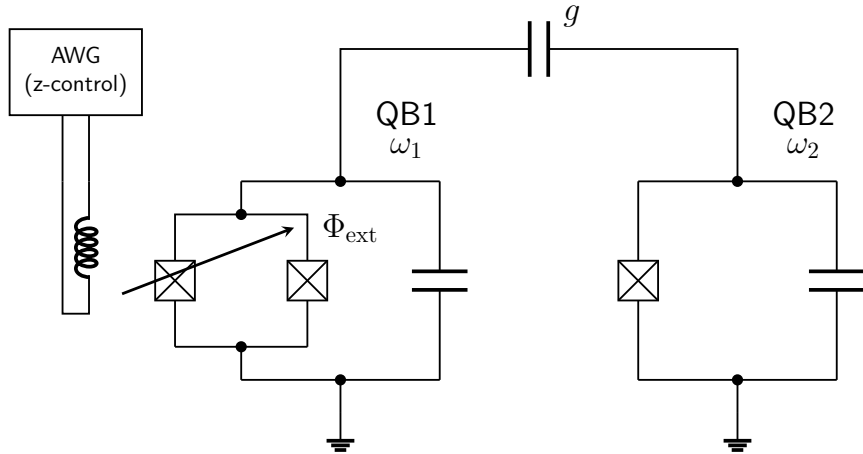


Figure 2: Target qubit architecture in our work. A tunable-frequency QB1 and a fixed-frequency QB2 are capacitively coupled with factor g .

In superconducting quantum architectures, the ability to dynamically tune the transition frequency of a qubit is essential for realizing the adiabatic CZ gate. This tunability is achieved by replacing the single Josephson junction of a standard transmon with a superconducting quantum interference device (SQUID) loop as depicted in Fig. 2, which is then threaded by a localized magnetic flux Φ_{ext} controlled via a dedicated microwave transmission line known as the Z -control line (or flux line).

3.2 Qubit Architecture

The system we consider is illustrated in Fig. 2. It is a system of two capacitively coupled transmon qubits, one fixed-frequency (QB2) and the other frequency tunable (QB1). The simplified dispersive Jaynes-Cummings Hamiltonian for this system can be modeled as:

$$H = \sum_{i=1,2} \left(\omega_i \hat{a}_i^\dagger \hat{a}_i + \frac{\alpha_i}{2} \hat{a}_i^\dagger \hat{a}_i^\dagger \hat{a}_i \hat{a}_i \right) + g(\hat{a}_1^\dagger + \hat{a}_1)(\hat{a}_2^\dagger + \hat{a}_2), \quad (19)$$

where g denotes the capacitive coupling strength between the two qubits, $\hat{a}_i^\dagger$ and $\hat{a}_i$ represent the creation and annihilation operators for qubit i , respectively, and α_i defines the anharmonicity of qubit i . Note that we use α for anharmonicity in discussion of the CZ gates. We use Δ and ω_i to denote different things. (Δ instead denotes the effective coupling strength $2\sqrt{2}g$)

3.3 Adiabatic Implementation

CZ gate is a two-qubit gate that applies a phase flip to the target qubit if the control qubit resides in the state $|1\rangle$. In the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, the ideal unitary evolution is represented by a diagonal matrix:

$$U_{CZ} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (20)$$

To implement this gate, the frequency of QB1 is swept along a prescribed trajectory $\omega_1(t)$ toward the avoided crossing, held or continuously varied for a duration τ_d , and then returned to its idle parking frequency. In this process, $|11\rangle$ obtains/loses more phases than the other computational states, due to its stronger coupling to $|02\rangle$.

By following Refs. [4, 23], we review the theory behind the adiabatic implementation of the CZ gate and the derivation of the leakage formula Eq. (25).

Theorem (Adiabatic Theorem). *If a quantum system is initially prepared in a non-degenerate energy eigenstate of a time-dependent Hamiltonian, and if that Hamiltonian is varied sufficiently slowly, the system will evolve continuously to*

remain in the corresponding instantaneous eigenstate of the evolving Hamiltonian up to an accumulated phase factor, given that there is a gap between the eigenvalue and the rest of the Hamiltonian's spectrum.

Consider a two level system with the following Hamiltonian

$$H(t) = \varepsilon(t)\sigma_z + \Delta\sigma_x.$$

According to the adiabatic theorem, if we prepare the state $|0\rangle$ and slowly vary the Hamiltonian and then retrace the change back, the resulting state is $|0\rangle$ with phase accumulation. The instantaneous eigenstates are

$$|\psi_-\rangle = \begin{bmatrix} -\sin(\theta(t)/2) \\ \cos(\theta(t)/2) \end{bmatrix} \quad \text{and} \quad |\psi_+\rangle = \begin{bmatrix} \cos(\theta(t)/2) \\ \sin(\theta(t)/2) \end{bmatrix}, \quad (21)$$

with eigenenergies

$$E_- = -\frac{1}{2}\sqrt{\varepsilon(t)^2 + \Delta^2} \quad \text{and} \quad E_+ = \frac{1}{2}\sqrt{\varepsilon(t)^2 + \Delta^2}, \quad (22)$$

where

$$\theta(t) = \arctan \frac{\Delta}{\varepsilon(t)}. \quad (23)$$

We denote the energy difference of these two instantaneous eigenstates as

$$\omega(t) = \sqrt{\varepsilon(t)^2 + \Delta^2}. \quad (24)$$

3.3.1 Calculation of Leakage Error

Consider a state

$$|\psi\rangle = \alpha |\psi_-\rangle + \beta |\psi_+\rangle = \alpha_0 |0\rangle + \beta_0 |1\rangle$$

with rotational change of coefficients

$$\begin{aligned} \alpha &= \alpha_0 \cos \frac{\theta}{2} + \beta_0 \sin \frac{\theta}{2}, \\ \beta &= \beta_0 \cos \frac{\theta}{2} - \alpha_0 \sin \frac{\theta}{2}. \end{aligned}$$

Differentiating with respect to time, we obtain

$$\begin{aligned} \dot{\alpha} &= \frac{-i\omega\alpha + \beta\dot{\theta}}{2}, \\ \dot{\beta} &= \frac{i\omega\beta - \alpha\dot{\theta}}{2}. \end{aligned}$$

By expressing the coefficients in terms of the angles $\alpha = \cos(\Theta/2)$ and $\beta = e^{i\phi} \sin(\Theta/2)$, we can define

$$\alpha^* \beta = \frac{\sin \Theta e^{i\phi}}{2}.$$

Calculating the time derivative of this quantity yields

$$\frac{d}{dt}(\sin \Theta e^{i\phi}) = i\omega \sin \Theta e^{i\phi} \pm \dot{\theta} \cos \Theta.$$

Plugging $\phi = \phi_{\text{initial}} + \int_0^t \omega(t') dt'$ and symplifying the expression gives the approximate leakage probability $P_e = |\sin(\Theta(T)/2)|^2$,

$$P_e \approx \frac{1}{4} \left| \int_0^T \frac{d\theta}{dt} e^{-i \int \omega(t') dt'} \right|^2, \quad (25)$$

with T being a gate duration.

3.3.2 Dynamical and Conditional Phase Accumulation

During the sweep towards the avoided crossing, computational states other than $|11\rangle$ also accumulate dynamical phases. To determine the net phase acquired by the $|11\rangle$ state, these single-qubit contributions must be subtracted. Specifically, the conditional phase ϕ_{cond} accumulated over the gate duration is calculated by isolating the two-qubit interaction phase from the single-qubit contributions,

$$\phi_{\text{cond}} = \Phi_{11} - \Phi_{01} - \Phi_{10} + \Phi_{00},$$

where Φ_{ij} denotes phase accumulation for the state $|ij\rangle$. A successful gate operation requires calibrating the pulse amplitude and duration such that $\phi_{\text{cond}} = \pi \pmod{2\pi}$.

3.3.3 Nonlinear time transformation

By applying a time transformation, we can simplify the leakage formula Eq. (25). Let the time variable of this new frame be τ . Set

$$\Delta d\tau = \omega(t)dt,$$

then

$$dt = \frac{\Delta}{\omega(t)} d\tau = \sin \theta(t) d\tau. \quad (26)$$

In integral form,

$$t(\tau) = \int_0^\tau \sin \tilde{\theta}(\tau') d\tau'. \quad (27)$$

The virtue of this transformation is clear if we plug Eq. (26) into Eq. (25),

$$P_e = \frac{1}{4} \left| \int_0^\tau \frac{d\tilde{\theta}}{d\tau} e^{-i\Delta\tau} d\tau \right|^2. \quad (28)$$

The leakage error is expressed as the Fourier transform of $\frac{d\tilde{\theta}}{d\tau}$ at Δ .

3.4 Slepian pulses

A (discrete) Slepian pulse is a finite-length discrete time signal which maximizes the energy concentration in a prescribed frequency band $[-W, W]$. The exact formula for this pulse can be derived by considering the following eigenvalue problem.

Consider a discrete-time pulse of finite length, denoted as $x[n]$ for $n = 0, 1, \dots, N-1$. Its energy concentration in $[-W, W]$ is

$$\frac{\int_{-W}^W |X(e^{i\omega})|^2 d\omega}{\int_{-\pi}^{\pi} |X(e^{i\omega})|^2 d\omega}, \quad (29)$$

where X denotes discrete Fourier transform,

$$X(e^{i\omega}) = \sum_{n=0}^{N-1} x[n] e^{-i\omega n}.$$

The Eq. (29) can be rewritten as

$$\frac{x^T A x}{x^T x}$$

where x is the vector form of the signal $x[n]$ and A is a matrix with $A_{n,m} = \frac{\sin 2\pi W(n-m)}{\pi(n-m)}$.

A has N eigenvalue solutions and the eigenvector corresponding to the second largest eigenvalue satisfies anti-symmetric property $x[n] = -x[N-1-n]$.

3.5 Limitations of Slepian Pulses

While the Slepian pulse is widely utilized and exhibits an excellent spectral profile with exceptionally low sidelobes, it is not strictly optimal for this specific application. As demonstrated in Eq. (28), the leakage error depends exclusively on the Fourier transform evaluated at the precise frequency Δ . Consequently, this leaves significant room for further targeted waveform optimization.

Chapter 4 Proposed Waveform Design Techniques

In this section, we propose methods to mitigate leakage errors in X and CZ gates, beginning with the X gate. Our approach builds upon the recursive DRAG schemes which conventionally rely on seed functions to generate a control pulse to minimize leakage caused by two photon transitions. However, this conventional method suffers from limited control over the resulting pulse shape. To address this, we establish a novel method to construct DRAG-1 pulses without relying on a seed function, thereby enabling shorter gate durations without compromising operational fidelity.

4.1 Inverse Design of the DRAG-1 Pulse

We now rewrite Eq. (14) as a second-order linear ordinary differential equation (ODE):

$$\Omega_x^2(t) = \left(1 + \frac{D^2}{\Delta_2^2}\right) \Omega_1^2(t). \quad (30)$$

Recognizing this as a driven harmonic oscillator equation, the general solution is

$$\Omega_1^2(t) = \int_0^t \left[\frac{1}{\Delta_2} \sin(\Delta_2(t - \tau)) \right] [\Delta_2^2 \Omega_x(\tau)^2] d\tau + A \cos(\Delta_2 t + \phi). \quad (31)$$

The first term represents the particular solution to the ODE in Eq. (30). To check the validity of this particular solution we use the Leibniz rule [1], which dictates the differentiation under the integral sign:

$$\frac{d}{dt} \int_{a(t)}^{b(t)} F(x, t) dx = F(b(t), t) \cdot b'(t) - F(a(t), t) \cdot a'(t) + \int_{a(t)}^{b(t)} \frac{\partial}{\partial t} F(x, t) dx. \quad (32)$$

Recall we have three boundary conditions on our debt:

1. $\Omega_x = 0$ at both ends ($t = 0, T$): This ensures compatibility with the SWT and enforces boundary continuity for the X-pulse, mitigating spectral leakage.
2. $\dot{\Omega}_x = 0$ at both ends ($t = 0, T$): This condition guarantees a smooth envelope derivative, maintaining essential boundary continuity for the corresponding Y-pulse.

3. $\Omega_1 = 0$ at both ends ($t = 0, T$): This further enforces strict SWT compatibility at the start and completion of the gate.

Applying the third boundary condition at the initial time $t = 0$ immediately eliminates the homogeneous component, yielding $A = 0$ in Eq. (31). We name

$$\Omega_1^2(t) = \int_0^t \left[\frac{1}{\Delta_2} \sin(\Delta_2(T - \tau)) \right] [\Delta_2^2 \Omega_x(\tau)^2] d\tau$$

the DRAG-1 formula in integral form. In the same parlance, we call Eq. (14) the differential DRAG-1 formula. Imposing the boundary constraint at $t = T$ yields

$$(\Omega_1^2(T)) = 0 = \int_0^T \left[\frac{1}{\Delta_2} \sin(\Delta_2(T - \tau)) \right] [\Delta_2^2 \Omega_x(\tau)^2] d\tau.$$

Differentiating both sides with respect to t , we obtain

$$(\Omega_1(T)\dot{\Omega}_1(T)) = 0 = \int_0^T [\cos(\Delta_2(T - \tau))] [\Delta_2^2 \Omega_x(\tau)^2] d\tau.$$

These two constraints combined implies

$$\int_{-\infty}^{\infty} [\Delta_2^2 \Omega_x(\tau)^2] e^{-i\Delta_2\tau} d\tau = 0.$$

Consequently, the boundary condition at $t = T$ translates to a null fourier transform condition

$$\int_0^T \Omega_x(\tau)^2 e^{-i\Delta_2\tau} d\tau = 0. \quad (33)$$

4.2 Inverse Design of the DRAG-2 Pulse

Recall that the DRAG-2 pulse in Eq. (16) has exactly the same form as the DRAG-1 in Eq. (14). Since the boundary conditions coming from the SWTs also have the same form ($\Omega_2(t) = 0$ at $t = 0, T$), we find that the existence of such a Ω_2 in Eq. (16) is guaranteed by

$$\int_0^T \Omega_1(\tau)^2 e^{-i\Delta_3\tau} d\tau = 0, \quad (34)$$

which is equivocal to $\mathcal{F}[\Omega_1](\Delta_3) = 0$. By taking Fourier transform of the both sides of the differential DRAG-1 formula Eq. (14), we obtain

$$\mathcal{F}[\Omega_x^2](z) = \left(1 - \frac{z^2}{\Delta_2^2}\right) \mathcal{F}[\Omega_1^2](z).$$

Hence, Eq. (34) is equivalent to $\mathcal{F}[\Omega_x](\Delta_3) = 0$.

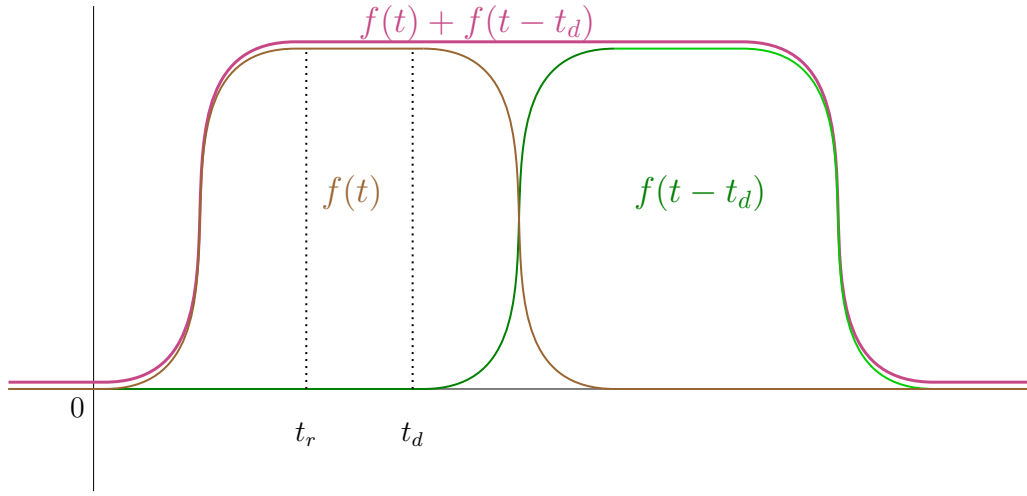


Figure 3: Creating a flat-top pulse using echo method

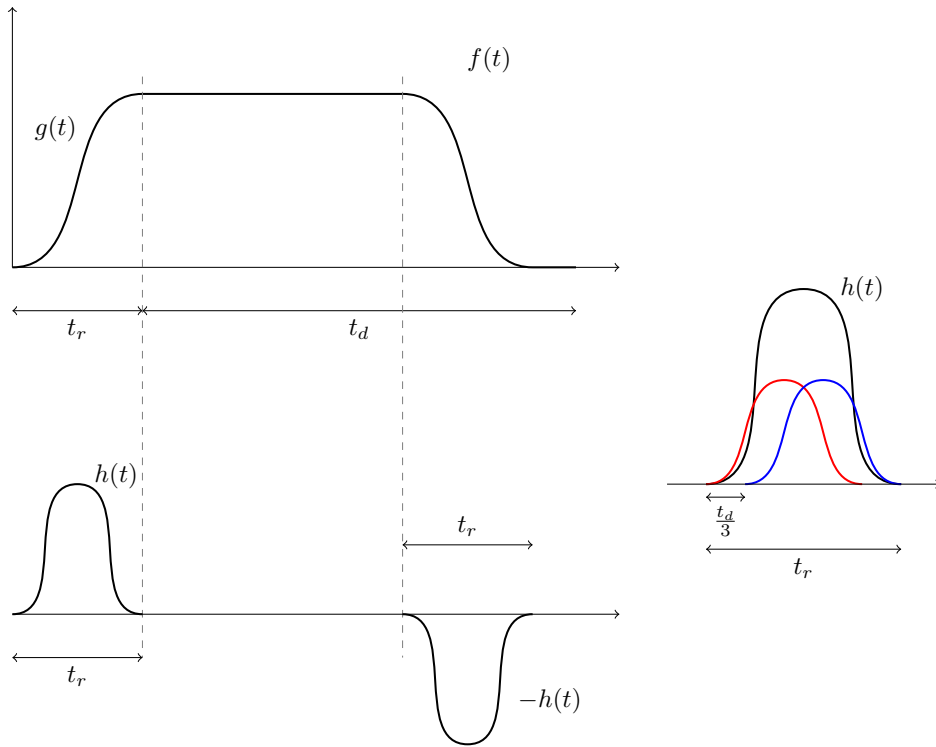


Figure 4: DRAG-2 method

4.3 Pulse Design

Having established that DRAG-1 and DRAG-2 formula are equivalent to zero Fourier transform at Δ and Δ_3 respectively, we now utilize the echo method to create pulses with these properties.

4.3.1 DRAG-1 pulse

We propose a novel method to synthesize a flat-top DRAG-1 pulse. Utilizing a flat-top envelope enables us to reduce the lower bound of the gate duration for which the Schrieffer-Wolff transformation (SWT) remains valid. This is because keeping the peak amplitude, $\max|\Omega_x|$, relatively small helps mitigate the inaccuracies introduced by the early truncation of the SWT series. To engineer a pulse that yields a zero Fourier transform at a specific target frequency, the echo method is a widely employed technique. This method is ubiquitous across numerous scientific and technological disciplines, including signal processing, nuclear magnetic resonance (NMR) [13], acoustics [24], and laser physics [22]. Its direct application to quantum gate operations has been previously explored in Ref. [26]. Although Jesus et al. [14] investigated optimal pulse waveforms by designing Ω_x indirectly via the auxiliary function Ω_1 , our methodology allows for the direct design of a DRAG-1 pulse using Eq. (33).

Let $t_d = \pi/\Delta$. For a given function $f(t)$, its delayed superposition $f(t) + f(t - t_d)$'s Fourier transform evaluated at Δ is

$$\mathcal{F}[f(t) + f(t - t_d)](\Delta) = \mathcal{F}[f](\Delta) + e^{i\Delta t_d} \mathcal{F}[f](\Delta) = 0.$$

We aim to design a pulse such that this sum results in a flat-top profile, as illustrated in Fig. 3. Here we also impose that $f(t)$ is a flat-top pulse, then the pulse design problem reduces to defining the waveform strictly on $[0, t_r]$, where t_r denotes the rise time.

By normalizing this rising edge, the objective can be restated as finding a function $g(t)$ on $[-1, 1]$ that satisfies the following property:

1. g is anti-symmetric.
2. g is monotonically increasing.
3. $g(1) = 1/2$.
4. The first and second derivative of g vanish at $t = -1$.

The first condition comes from the additional constraint that the $f(t)$ is also a flat-top pulse. The second condition is set so that it maximally reduces gate duration. The fourth condition ensures that $\Omega_x = \dot{\Omega} = 0$ are satisfied at $t = 0, T$. Once obtaining such a g , we reconstruct Ω_x^2 as

$$\Omega_x^2(t) = \begin{cases} A(g(2t/t_r - 1) - g(-1)) & (0 < t < t_r) \\ A & (t_r < t < t_d) \\ A(g(2(t_r + t_d - t)/t_r - 1) - g(-1)) & (t_d < t < t_d + t_r) \end{cases} \quad (35)$$

where A is the peak amplitude, determined by integrating Ω_x^2 according to Eq. (15).

4.3.2 DRAG-2 pulse

We now extend the method described above and gain a pulse which also satisfies the DRAG-2 formula. Note that for a function $f(t)$,

$$iz\mathcal{F}[f](z) = \mathcal{F}[f'](z).$$

This implies that the requirement that Ω_x^2 have a zero Fourier transform at Δ_3 is equivalent to its derivate have a zero Fourier transform at Δ_3 . If $h(t)$ is the derivative of $g(t)$, $h(t) = g'(t)$, the derivative of the flat-top pulse in 4.3.1 consists of two $h(t)$ and $-h(t - T + t_r)$ as depicted in Fig. 4. $h(t)$ is a symmetric function since it is a derivative of a anti-symmetric function.

We can create $h(t)$ so that it has zero Fourier transform at Δ_3 using the echo method as illustrated in Fig. 4. Theoretically, $\Delta_n = \frac{n(n-1)}{2}\Delta$, so the delay time to cancel the Fourier transform is

$$\frac{\pi}{\Delta_3} = \frac{\pi}{3\Delta} = \frac{t_d}{3}.$$

4.4 Optimization via Riccati Engineering

As we've seen in 2.7.1, NPAD has an obvious issue that the Hamiltonian is assumed to be time-independent, and in Ref. [20], the author only applies this method to perturbative cases, in which the method reduces to recursive Schrieffer-Wolff transformations.

We outline the newly proposed procedure to handle non-perturbative cases.

Applying to the qudit Hamiltonian in Eq. (6) with $N_{\max} = 2$, the Riccati equation becomes

$$\begin{aligned} i\dot{x}_1 &= (2\delta + \Delta)x_1 - \frac{1}{2}(\Omega_x + i\Omega_y)x_2 - \frac{\lambda}{2}(\Omega_x - i\Omega_y)x_1x_2, \\ i\dot{x}_2 &= \frac{\lambda}{2}(\Omega_x + i\Omega_y) + (2\delta + \Delta)x_2 - \frac{1}{2}(\Omega_x - i\Omega_y)x_1 - \frac{\lambda}{2}(\Omega_x - i\Omega_y)x_2^2. \end{aligned} \quad (36)$$

If we define

$$\begin{aligned} A &= -\frac{1}{2}(x_2^2 + \lambda x_1), \\ B &= \frac{1}{2}x_1^2, \\ C &= i(x_1x_2 - x_1\dot{x}_2). \end{aligned} \quad (37)$$

Then Ω_x , Ω_y , and δ can be expressed as

$$\tilde{\Omega} = \Omega_x + i\Omega_y = \frac{A^*C - BC^*}{|A|^2 - |B|^2}, \quad (38)$$

$$\delta(t) = \frac{1}{2} \left(\frac{i\dot{x}_1 + \frac{1}{2}x_2\tilde{\Omega} + \frac{1}{2}\lambda x_1x_2\tilde{\Omega}^*}{x_1} - \Delta_2 \right). \quad (39)$$

4.4.1 Variational Pulse Adjustment

A direct approach to applying Eq. (38) and Eq. (39) involves finding x_1 and x_2 that lands on 0 at $t = 0, T$, and the time evolution given by Eq. (18) corresponds to a X gate. At the very end of this process, we find a numerical solution for $\tilde{\Omega}$ and δ . However, this methodology introduces several analytical and numerical challenges. Most notably, the non-linearity of the Riccati equation Eq. (17) substantially complicates the precise calibration of gate operations. To circumvent this issue, we reframe the pulse shaping problem within a continuous optimization framework utilizing a variational method. From existing $\tilde{\Omega}$ and δ , we numerically solve the corresponding x_1 and x_2 , then we add variations to thus obtained x_1 and x_2 to reduce leakage errors and achieve higher fidelity operations. To preserve the X gate condition Eq. (15),

$$\delta\Theta = \int_0^T \sum_{k=1}^2 \left(\frac{\partial\Omega_x}{\partial x_k} \delta x_k + \frac{\partial\Omega_x}{\partial \dot{x}_k} \delta \dot{x}_k + \text{c.c.} \right) dt \quad (40)$$

must vanish. ($\frac{\partial}{\partial x_i}$ denotes Wirtinger derivatives.) Integral by parts transforms the above equation into

$$\delta\Theta = \left[\sum_{k=1}^2 (P_k(T)\delta x_k(T) + P_k^*(T)\delta x_k^*(T)) \right] + \int_0^T \sum_{k=1}^2 (G_k(t)\delta x_k(t) + G_k^*(t)\delta x_k^*(t)) dt, \quad (41)$$

where $G_k(t)$ and $P_k(t)$ ($k = 1, 2$) are defined as

$$\begin{aligned} G_k(t) &= \frac{\partial \Omega_x}{\partial x_k} - \frac{d}{dt} \left(\frac{\partial \Omega_x}{\partial \dot{x}_k} \right), \\ P_1(t) &= \frac{\partial \Omega_x}{\partial \dot{x}_1} = i \frac{A^* - B^*}{2(|A|^2 - |B|^2)} x_2, \\ P_2(t) &= \frac{\partial \Omega_x}{\partial \dot{x}_2} = -i \frac{A^* - B^*}{2(|A|^2 - |B|^2)} x_1. \end{aligned}$$

Note that we have $\delta x_k(0) = 0$ since we already have $x_k(0) = 0$ as an initial value of the integration.

To mitigate the coupling of computational and non-computational subspaces, we want $x_1(T)$ and $x_2(T)$ as small as possible.

Ideally, we can totally cancel the leakage error by setting $\delta x_k(T) = -x_k(T)$. However, since this is a variational method, $\delta x_k(t)$ should be sufficiently small and in practice, we multiply some small number ϵ to thus obtained $x_k(t)$.

4.5 CZ gate

In this section, for brevity, the length of pulses are assumed to be N , an odd number. As seen in 3.3.1, if we denote with g the derivative of θ we design in non-linear time frame, the leakage error is approximately expressed as $\frac{1}{4}|\mathcal{F}[g](\Delta)|^2$. Although both formulations (X gate and CZ gate) share the similar objective of minimizing or zeroing out the Fourier transform, the specific methods we employ for each gate differ. This distinction arises because, in practical applications of the CZ gate, it is more critical to maintain a small Fourier transform amplitude over an interval of frequencies [23].

In the following, we propose three methods to achieve this. One is a proposal of a novel pulse shape. The other two are optimization techniques with sequential quadratic programming.

4.5.1 The Anti-symmetric Chebyshev Pulse

Maintaining a specified sidelobe level, Chebyshev pulse is the signal that minimizes the mainlobe width.

Mathematically, a chebyshev pulse g_{ch} is a discrete time pulse defined as

$$g_{ch}[n] = \frac{1}{N} \left[1 + 2r \sum_{k=0}^{(N-1)/2} (-1)^k T_{N-1} \left(x_0 \cos \frac{\pi k}{N} \right) \cos \left(\frac{2\pi k}{L} \left(n + \frac{1}{2} \right) \right) \right],$$

where $r = \frac{1}{T_{N-1}(x_0)}$ corresponds to the suppressed sidelobe level. We call its anti-symmetrization

$$g_{ach}[n] = \begin{cases} g_{ch}[n] & n < (N+1)/2 \\ 0 & n = (N+1)/2, \\ -g_{ch}[N-n] & n > (N+1)/2 \end{cases}$$

anti-symmetric Chebyshev pulse g_{ach} .

4.5.2 Spectrum Suppression via Sequential Quadratic Programming

There are many methods known to tailor a pulse with a target spectrum. Weighted Chebyshev approximation (WCA) [4] is one of them. In WCA, you have two prescribed parameters, a weight function $W(w)$ and a spectrum profile $A(w)$. WCA is a method to find a pulse whose spectrum is $G(w)$ such that the maximum L^∞ error

$$\max\{|W(w)(G(w) - A(w))|\}$$

is minimized. WCA has the following disadvantages.

- **over-constraining** Although the goal is to optimize the “amplitude” (intensity) of the spectrum while the phase is left arbitrary, because the WCA objective function $\max\{|W(w)(G(w) - A(w))|\}$ is evaluated in the complex domain, we need to design “good” choice of phase beforehand, making the implementation of algorithm difficult.
- **Outliers** By the very nature of Chebyshev approximation, WCA seeks to minimize the “maximum error” across the entire evaluation function (minimizing the $\mathcal{L}_\infty$ norm). This means that if the spectrum deviates significantly from the target at even a single point (such as a sharp peak or a minor dip), the algorithm will focus excessively on correcting that specific

outlier, risking an overall distorted pulse waveform.

In the following, we present two formulations of our optimization problem. We formulate them in a form solvable in the framework of Sequential Quadratic Programming [27].

4.5.3 Redefining the Slepian eigenvalue problem

Recall from 3.4, the (second) Slepian pulse is the solution of the second largest eigenvalue to the following problem

$$\frac{\int_{-W}^W |G_{\text{DTFT}}(\omega)|^2 d\omega}{\int_{-\pi}^{\pi} |G_{\text{DTFT}}(\omega)|^2 d\omega} = \frac{g^T A_{m,n} g}{g^T g} \quad (42)$$

where $A_{m,n} = 2 \text{sinc}((m-n)W) = \frac{\sin(2\pi(m-n)W)}{\pi(m-n)W}$, the Fourier transform of a rectangular function. (42) is the eigenvalue problem of the matrix $A_{m,n}$. It was Slepian who found a commuting matrix for $A_{m,n}$ and solve the problem analytically.

Suppose our target range of the gate duration satisfies For the target range of $W_1 \leq \frac{T}{N}\Delta \leq W_2$. Then the original maximization problem turns into the following minimization problem:

$$\text{minimize } \frac{\int_{W_1}^{W_2} |G_{\text{DTFT}}(\omega)|^2 d\omega}{\int_{W_1}^{W_2} |G_{\text{DTFT}}(\omega)|^2 d\omega} = \frac{\tilde{g}^T (A_{\text{rect}}(W_2) - A_{\text{rect}}(W_1)) \tilde{g}}{\tilde{g}[0]^2 + \dots + \tilde{g}[N]^2}$$

The conventional use of Slepian pulse is quite casual about the target gate time range of operation, and Slepian pulse is the best pulse if the above W_2 is set as $W_2 = \pi$.

The difficulty arising by reformulating the problem in this fashion is the problem is no longer analytically solvable. To clarify the search space of g , we impose the following conditions: To not elongate gate time we impose the following condition,

- $g[s] = g[N-1-s]$: (anti-symmetry) To mitigate $1/f$ noise.
- $g[s] \geq 0$ for $0 \leq s \leq L-1$: (positivity) To not elongate gate duration.

If we define,

$$B = \begin{pmatrix} 1 & & & \\ & \ddots & & \\ & & 1 & \\ 0 & \dots & 0 & \\ & & & -1 \\ & & \ddots & \\ -1 & & & \end{pmatrix}, \quad C = B^T A B,$$

$$\begin{pmatrix} \tilde{g}[0] \\ \vdots \\ \tilde{g}[N] \end{pmatrix} = B \begin{pmatrix} h[0] \\ \vdots \\ h[L-1] \end{pmatrix}.$$

Then

$$\text{maximize} \quad -\frac{h^T C h}{h^T h},$$

with h being a positive vector of length $\lfloor N \rfloor / 2$ is the problem to solve.

4.5.4 Sidelobe-Aware Optimization

Unlike the formulation given above, the second formulation is more of a heuristic sort, more focused on the initial pulse shape f . We find its sidelobe boundaries first and impose different allowed gain levels for each distinct sidelobe, suppressing the targeted frequency band while relaxing constraints elsewhere to minimize time-domain distortion. Since we want to stick to these sidelobes, we set the objective function to be the square sum of the difference from the initial pulse shape.

Chapter 5 Simulation and Numerical Results

For quantum simulation, we used QuTiP5 [18] software. We first see the results from X gate simulation.

5.1 X gate

Simulation has been conducted with the Hamiltonian in Eq. (7) with $N_{max} = 3$. Among several solvers that QuTiP provides, the Shroedinger solver has been chosen. This is enough unless your target fidelity level is exceedingly low so that the effect of decoherence is visible in the duration of one gate operation.

The parameters we use for this simulation are

- $\Delta/2\pi = -0.225$ GHz (anharmonicity),
- $\lambda_k = \sqrt{k}$.

We set the time slice number of the ODE solver as 25000. (Increasing this number didn't change the simulation results.)

5.1.1 Fidelity

The gate infidelity was computed as $\mathcal{E}(\hat{U}_Q) = 1 - \mathcal{F}[\hat{U}_Q]$ where $\mathcal{F}[\hat{U}_Q]$ is the gate fidelity defined as [25]

$$\mathcal{F}[\hat{U}_Q] = \frac{\text{Tr}[\hat{U}_Q \hat{U}_Q^\dagger]}{d(d+1)} + \frac{|\text{Tr}[\hat{U}_Q \hat{U}_I^\dagger]|^2}{d(d+1)}. \quad (43)$$

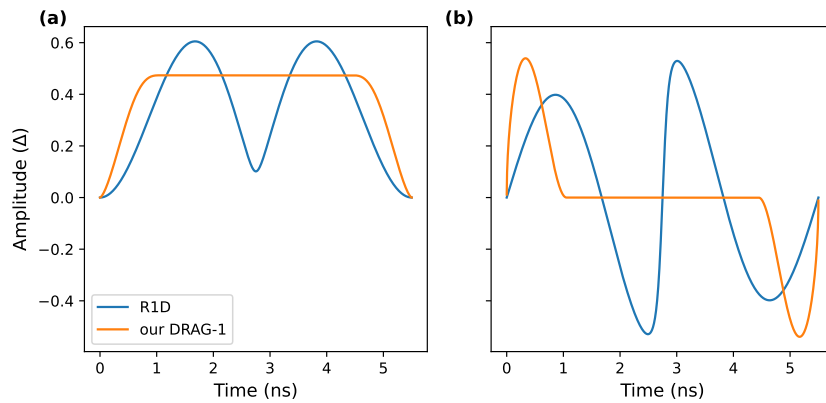


Figure 5: Comparison of X and Y controls of our DRAG-1 (orange) and R1D (blue) pulses. (a) X control, (b) Y control.

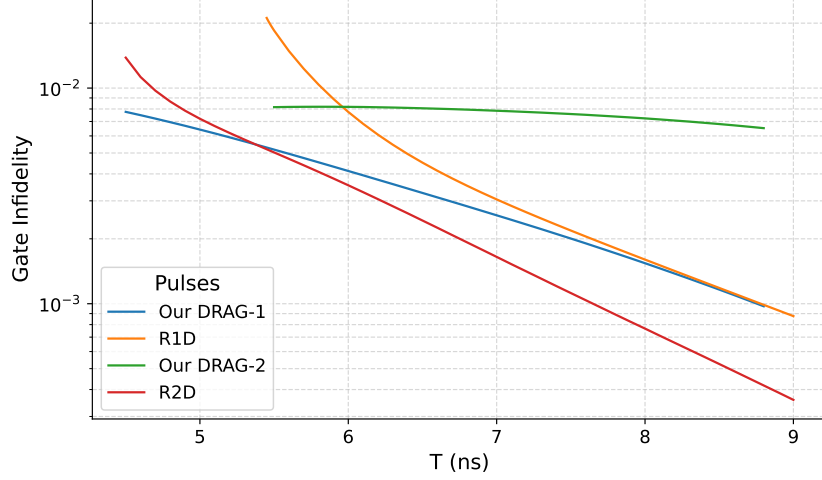


Figure 6: Although our DRAG-2 pulse shows poor performance, our DRAG-1 pulse surpasses the performance of the R2D pulse for $T < 5.3$ ns.

5.1.2 Pulse shapes

In our simulaiton setting, $t_d = 2.2$ ns.

DRAG-1 Pulse

For this analysis, two variations of DRAG-1 pulses were synthesized. The first was derived via the standard differential method using the seed function $\sin^3(\pi t/T)$. The second was constructed using our proposed methodology, which is characterized by a rising edge function of the form $g(t) = x + \sin(x)$. Throughout the remainder of this text, the pulse based on the $\sin^3(\pi t/T)$ seed is designated as R1D, while the pulse generated by our novel approach is referred to as our DRAG-1.

DRAG-2 Pulse

In [21], the pulse shape

$$\frac{1}{16} \cos(6\pi) - \frac{9}{16} \cos\left(\pi \frac{2t}{t_f}\right) + \frac{1}{2}.$$

is a versatile initial pulse shape for recursive DRAG framework. We call R2D for the DRAG-2 pulse created from this pulse.

With the function $h(t)$ being a raised cosine function, we curated our DRAG-2 pulse. The pulse shape for $T = 5.5$ ns is shown in Fig. 5.1.5.

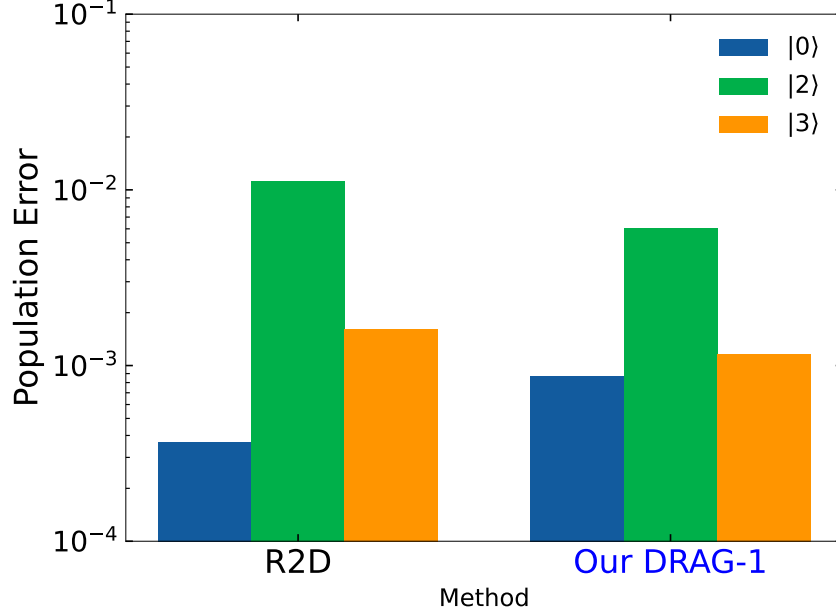


Figure 7: Population distribution of X gate for $T = 4.8$ ns implemented by R2D and our DRAG-1 pulse. The main error comes from the leakage to $|2\rangle$ in both cases.

5.1.3 Performance

In Fig. 5.1.1, we compare the gate infidelity between these four pulses. DRAG-1 outperformed and showed its effectiveness within short gate durations (< 5 ns). As the gate duration decreases, both R1D and R2D exhibit a non-linear degradation in performance. This is attributed to the increased pulse amplitude, which compromises the accuracy of the SWT approximation. Compared to R1D and R2D, our DRAG-1 is specifically designed to suppress this amplitude growth. Consequently, its gate infidelity-gate duration characteristic remains consistent regardless of reductions in gate time.

5.1.4 Minimum Gate Time of Our DRAG-1

Using the echo method in the construction of the DRAG-1 pulse, as in our proposed method, has the minimum gate time limit

$$T = t_r + 2t_d \geq 2t_d \approx 4.44 \text{ ns.}$$

The minimum gate time of the Hann-type pulse is known to be $\sqrt{6}t_d \approx 5.44$. Hence, our DRAG-1 pulse has reduced 20% of gate duration compared to the

R1D pulse.

Conventional transmons anharmonicity ranges from -150 to -300 GHz [16], corresponding to $t_d = 1.7$ ns to 3.3 ns. Hence, the minimum gate duration of our method when applied to different transmon qubits ranges from 3.4 ns to 6.6 ns.

5.1.5 Limitations on our proposed design of DRAG-2 Pulse from the Echo Method

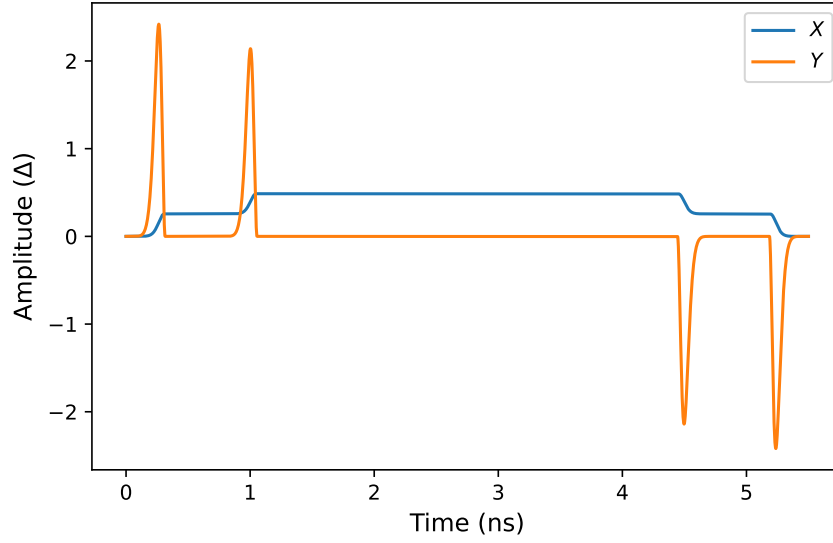


Figure 8: DRAG-2 pulse for $T = 5.5$ ns. $t_r < \frac{2}{3}t_d$ degrades the gate performance and sharp edges arise in the Y control.

In our proposed DRAG-2 design scheme, using the echo method in the construction of the rising edge of $f(t)$ in Fig. 3 carries several difficulties. First, it imposes us a minimum gate time limit as $t_r > t_d/3$,

$$T = 2t_d + t_r > 7/3t_d \quad \text{Our method.}$$

If we could use a different method to construct $f(t)$ in Fig. 3, this limit can be pushed to

$$T = 2t_d + t_r > 5/2t_d \quad \text{Theoretical limit.}$$

The minimum gate duration degrades by 7%. It has another problem of introducing edges into the Y control pulse as illustrated in Fig. 5.1.5. This is because

if $t_r < 2/3t_d$, $h(t)$ in Fig. 4 becomes short in duration. Although designed as a DRAG-2 pulse, and inherently performs better than DRAG-1,

5.1.6 Non-perturbative Validation of the Y-DRAG Correction

Aside from the X control pulse, we utilize the DRAG pulses 11 and 12. We demonstrate that the analytical expression for the Y control pulse remains valid even in the non-perturbative regime—a derivation that, to the best of our knowledge, has not been previously reported in the literature. To illustrate this, we recalculate the system Hamiltonian without relying on perturbative assumptions. We introduce a unitary frame transformation V , defined as:

$$V = \exp \left[-i\lambda_2 \frac{\Omega_x}{2\Delta_2} \hat{\sigma}_{1,2}^y \right] = \cos(\phi) - i\hat{\sigma}_{1,2}^y \sin(\phi) \quad (44)$$

where $\phi = \lambda_2 \frac{\Omega_x}{2\Delta_2}$, and apply it to our qudit Hamiltonian in Eq. (7). The $|1\rangle\langle 2|$ component of the resulting Hamiltonian is

$$-\sin(2\phi) \frac{\delta(t) + \Delta}{2} + \cos(2\phi) \frac{\lambda\Omega_x}{2} - i\frac{\lambda}{2} \left(\frac{\Omega'_x(t)}{\Delta} + \Omega_y(t) \right).$$

Substituting Ω_y as defined in Eq. (11) demonstrates that this off-diagonal component is still effectively suppressed. Consequently, we retain the standard DRAG correction for the Y control pulse.

5.1.7 Hardware Constraint

The proposed method utilizes a fast-rising flat-top pulse. We now evaluate the specifications such a pulse demands from the control hardware. To determine whether an Arbitrary Waveform Generator (AWG) can physically synthesize it, we must account for real-world hardware constraints—most notably the analog bandwidth limit and the slew rate limit. For this analysis, we assess these requirements against the specifications of the following AWGs, which are widely deployed in modern quantum computing architectures [28]:

- Qblox QCM (1GSa/S, BW 400MHz)
- Zurich Instruments SHFQC+ (2 GSa/s, 800 MHz)
- Keysight M8195A (65 GSa/s, 25 GHz)

Furthermore, the target drive amplitude Ω_x and the physical AWG output voltage V_{AWG} are not equivalent. They are related by a system-specific calibra-

tion factor c :

$$\Omega_x = cV_{AWG}.$$

While the exact value of c varies across different quantum processors, we assume $c = 50$ MHz/V for this evaluation. If the maximum AWG output is insufficient to reach the desired pulse amplitude under this conversion factor, a room-temperature low-noise RF amplifier must be incorporated in the control line.

The Analog Bandwidth Limit

By substituting $T = 5.0$ ns and $t_d = 2.2$ ns in $t_r + 2t_d = T$, we have the estimate $t_r = 0.6$ ns. By applying the RC Low-Pass Rise Time Formula [11], the rise time limit is $\frac{0.35}{BW}$, which is 0.43 ns for SHFQC+ and 0.86 ns for QCM.

Sampling Rate

An inadequate sampling rate at the rising edge will distort the synthesized pulse shape. If we assume a minimum of 5 to 6 samples is needed for accurate reproduction, the required rate is:

$$\frac{1}{0.6 \text{ ns}} \times 5 = 8.3 \text{ GS/s}$$

While only the Keysight M8195A meets this threshold among the evaluated models, an AWG of this high grade is not strictly mandatory.

The Slew Rate Limit

Inspecting Fig. 5.1.1, the amplitude can be estimated as 0.5Δ , which yields:

$$\frac{0.5\Delta}{c} \approx 4.0 \text{ V}$$

This indicates that the AWG alone, which generally outputs 1 to 2 V peak-to-peak, cannot generate the required voltage. Therefore, a lab-grade external amplifier is required, though this remains a highly practical solution.

5.2 Riccati Engineering

As for δx_k , we used the sum of two sin waves so that they the RHS of Eq. (41) becomes 0. Riccati engineering method generally failed to improve gate fidelity. This is considered to be due to difficulty in numerical analysis.

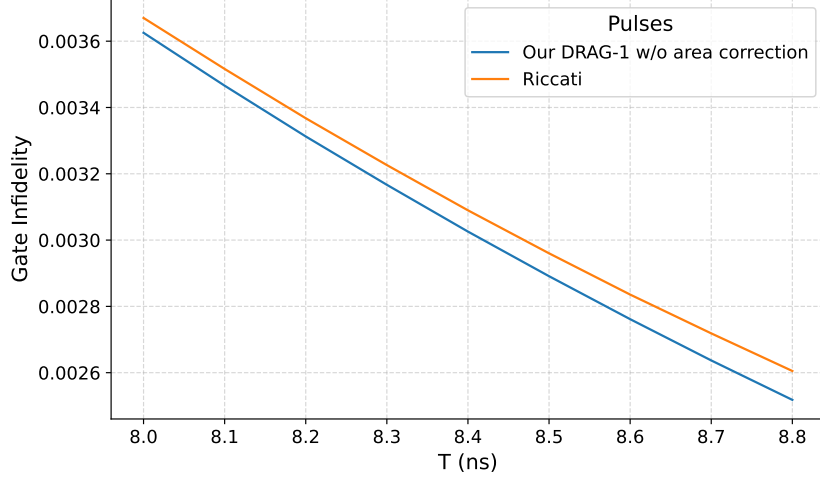


Figure 9: Our DRAG-1 pulses' performance when Riccati engineering applied. Riccati engineering had a little effect on gate infidelity.

5.3 CZ gate

5.3.1 Simulation

We set $N = 1001$. As in the previous research [4, 23], we simulate our designed pulses in the following fashion: We design $\frac{d\theta}{d\tau}$ and then integrate to find θ in non-linear time frame. We apply the time transformation to obtain $\theta(t)$. We then simulate the time evolution as the two-level system described in 3.3.1 to find the leakage error. We also simulate with the Hamiltonian in Eq. (19) to find the phase accumulation of $|00\rangle$, $|01\rangle$ and $|10\rangle$. We first simulate with the normalized amplitude, changing this amplitude by an optimizer to find when the conditional phase is close to π . Also, Slepian pulse with $NW = 2.9$ is used for reference.

5.3.2 Anti-symmetric Chebyshev pulse

Created with the parameter $r = 0.000238$, this pulse is illustrated in Fig. 10

5.3.3 Sidelobe-Aware Optimization

After the first four sidelobe, from 5-th to 13-th sidelobe heights are suppressed. The pulse shape and its spectrum are illustrated in Fig. 11.

5.3.4 Slepian pulse with redefined maximization problem

We set $W_1 = 2.9/1001$ and $W_2 = 8.8/1001$ to obtain a pulse with the corresponding redefined problem.

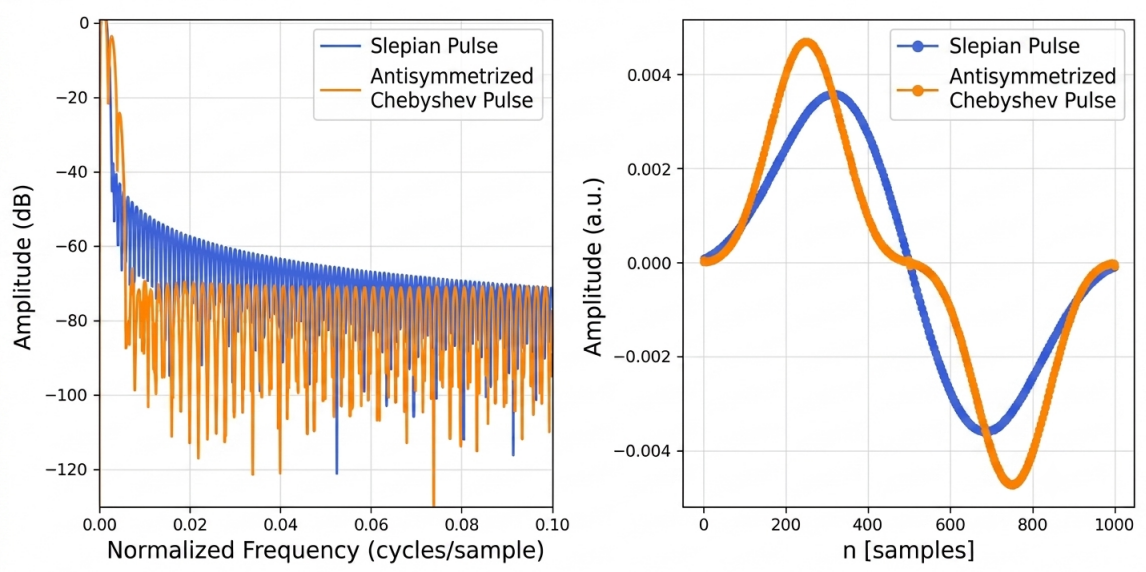


Figure 10: The frequency domain (left) and the time domain (right) representation of Slepian and antisymmetrized Chebyshev pulses. The parameter r is chosen so that sidelobe height is 40 dB lower than the first slepian sidelobe height.

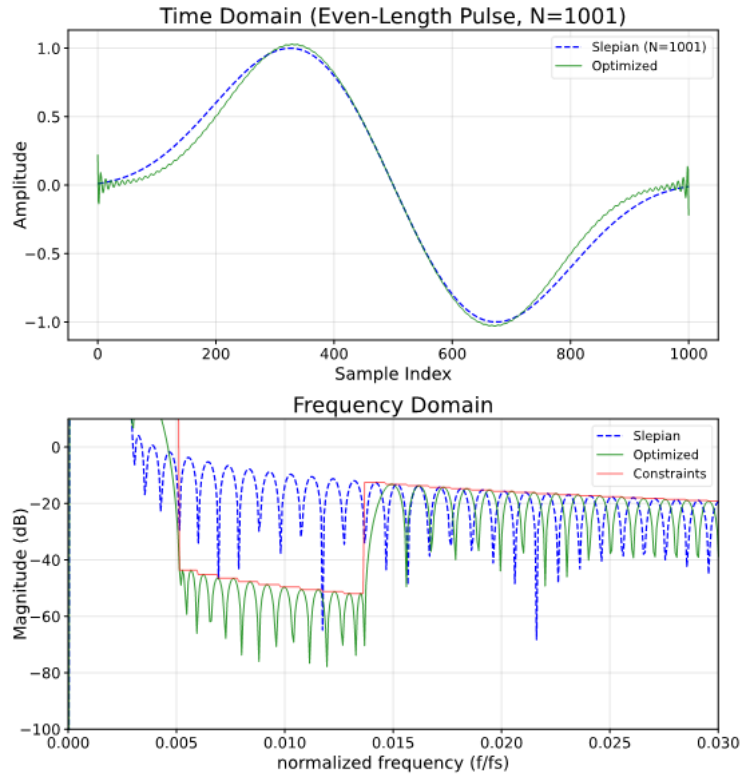


Figure 11: Slepian pulse and its optimized pulse.

5.3.5 Performance

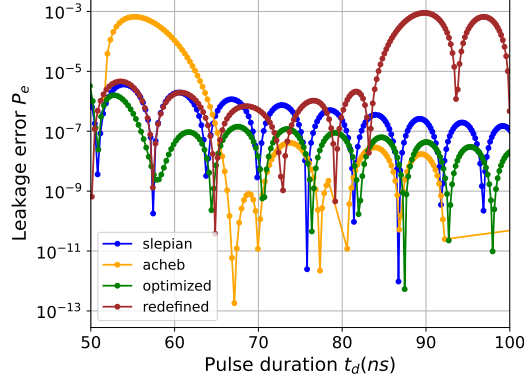


Figure 12: Leakage error of four pulses, Slepian (slepian), Anti-symmetric Chebyshev (acheb), Sidelobe-aware optimization (optimized), and the pulse optimized by redefining the original eigenvalue problem (redefined)

As illustrated in Fig. 12, in overall, the optimized slepian pulse performed the best. The anti-symmetric Chebyshev pulse performed really well at around $T = 70$ ns with suppressing leakage level by an order of two. The result of the Slepian pulse with the redefined problem is aligned with our expectations, larger leakage error at unwanted gate duration times.

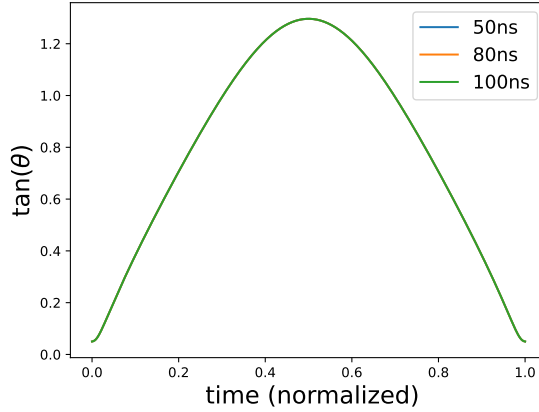


Figure 13: The relation between T_d and $\tan(\theta)$ for the Slepian pulse with $T = 50, 80, 100$. The three graphs are overlapping each other due to almost the same $A(\tau_d)$.

5.3.6 Minimum Gate time

Once the pulse shape g determined, its integration θ is decided up to the amplitude factor $A(\tau_d)$, ($0 \leq A(\tau_d) \leq 1$)

$$\theta = \theta^n(\tau/\tau_d)A(\tau_d),$$

where θ^n is the normalized θ that reaches $\epsilon(t_{mid}) = 0$ during avoided crossing.

$$\begin{aligned} t(\tau) &= \int_0^\tau \sin \theta(\tau') d\tau' \\ &= \tau_d \int_0^{\tau/\tau_d} \sin [A(\tau_d)\theta^n(\tau')] d\tau'. \end{aligned}$$

Therefore, gate duration in the lab frame is $t_d = \tau_d \int_0^1 \sin [A(\tau_d)\theta^n(\tau')] d\tau'$. t_d is decided upon the amplitude $A(\tau_d)$ and the initial pulse shape. As you see in Fig. 13, $A(\tau_d)$ does not change over different τ_d . Hence the gate durations in t and τ are proportionate.

Chapter 6 Conclusion

In this thesis, we have explored advanced gate pulse waveform optimization techniques for superconducting qubits, aiming to suppress errors and improve fidelity across both single-qubit and two-qubit operations. By systematically analyzing and overcoming the limitations of conventional pulse shaping methods, this research introduces novel, highly efficient pulse generation strategies.

The primary findings and methodological advancements of this work are summarized below:

Optimization of Single-Qubit X Gates

Mathematical Reformulation of recursive DRAG: We mathematically reformulated the standard DRAG-1 and DRAG-2 frameworks to invent a novel pulse design methodology.

Short Gate-Time Efficacy: The proposed method demonstrated significant effectiveness, particularly in minimizing errors at short gate times.

Exploration of Advanced Control Techniques: While the integration of advanced techniques such as Riccati engineering did not yield a drastic performance improvement in this specific configuration, combining these theoretical frameworks provided valuable insights into the fundamental limits of analytical pulse design.

Development of Two-Qubit CZ Gates

Introduction of the anti-symmetrized Chebyshev Pulse: As an alternative to the conventional Slepian pulse, we confirmed the performance of this novel pulse. This pulse is more suited if the gate duration is longer and anti-symmetrized Chebyshev pulses is a better alternative if the gate duration ranges greatly.

Introduction of a novel versatil optimization method: We investigated the performance bottleneck observed when using the Weighted Chebyshev Approximation (WCA). The conventional WCA approach yielded results that did not differ significantly from standard Slepian pulses. However, by applying Sequential Quadratic Programming (SQP) as an alternative optimization method, we achieved distinct improvements.

6.1 Future Work

Although this study utilized the echo method to synthesize the DRAG-2 pulse, alternative techniques for generating pulses with a zero Fourier transform at targeted frequencies warrant further investigation. Furthermore, future work will focus on calculating the DRAG-2 pulse from our proposed pulse, refining the integration of Riccati engineering to better leverage its potential in non-linear quantum control, and extending the generalized Slepian optimization and sidelobe-aware optimization techniques to address a broader range of pulse optimization problems within the field.

Acknowledgments

I would like to express my gratitude to Professor Takashi Sato for introducing me to the realm of quantum information and quantum control, and for providing the opportunity to undertake this research.

My deep thanks also go to Professor Kiichi Niitsu, Professor Masanori Hashimoto, Associate Professor Rei Ueno, Assistant Professor Kotaro Matsuka, and Professor Hiromitsu Awano (Nagoya University) for their valuable support through research workshops and seminars.

I am uniquely indebted to Ms. Shuko Nishiyama of the Sato Laboratory support staff for her continuous support of my day-to-day research activities. Finally, I extend my sincere appreciation to all members of the Integrated Systems Engineering Laboratory for creating such a vibrant and motivating research atmosphere.

Additionally, all numerical calculations and open quantum system simulations in this study were performed using the QuTiP 5 software library in Python. The author also thanks Wolfram Research, Inc. for the use of Mathematica (Version 14.3.0) in conducting the analytical calculations.

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Appendix

A.1 SWTs to Derive DRAG-2 Formula

We successively apply the following SWTs:

$$\begin{aligned}
\hat{S}_4(t) &= -i\lambda_2\lambda_3 \frac{\Omega_V^2(\Delta_3 - 2\Delta_2)}{8\Delta_3\Delta_2^2} \hat{\sigma}_{1,3}^y, \\
\hat{S}_5(t) &= i\lambda_2\lambda_3 \frac{(\dot{\Omega}_V\Omega_V)(\Delta_3 - 2\Delta_2)}{(8\Delta_3^2\Delta_2^2)} \hat{\sigma}_{1,3}^x, \\
\hat{S}_6(t) &= -i\lambda_2\lambda_3 \frac{\Omega_2^2(2\Delta_2 - \Delta_3)(\Delta_3^2 - \Delta_2^2)}{(8\Delta_2^4\Delta_3)} \hat{\sigma}_{1,3}^y, \\
\hat{S}_7(t) &= i\lambda_2\lambda_3 \frac{\dot{\Omega}_2\Omega_2(2\Delta_2 - \Delta_3)(\Delta_3^2 - \Delta_2^2)}{(4\Delta_2^4\Delta_3^2)} \hat{\sigma}_{1,3}^x.
\end{aligned}$$

A.2 Wolfram Code for Validating Y DRAG Correction without Perturbation

```

1 ClearAll["Global`*"];
2 M = 3;
3 dim = M + 1;
4
5 ketbra[m_, n_] := Normal@SparseArray[{{m + 1, n + 1} -> 1}, {dim,
   dim}]
6
7 (* Define the Pauli operators acting on adjacent levels |j-1> and |j>
   *)
8 sigX[j_] := ketbra[j - 1, j] + ketbra[j, j - 1]
9 sigY[j_] := -I * ketbra[j - 1, j] + I * ketbra[j, j - 1]
10 sigZ[j_] := ketbra[j - 1, j - 1] - ketbra[j, j]
11 comm[A_, B_] := A . B - B . A
12
13 (* =====
14    Define the Original Hamiltonian & Generator
15    ===== *)
16 H0 = Sum[(Delta[j] + j * delta[t]) * ketbra[j, j], {j, 1, M}];
17
18 Vx=Sum[lambda[j] * Omegax[t] / 2 * sigX[j], {j, 1, M}];
19 Vy=Sum[lambda[j] * Omegay[t] / 2 * sigY[j], {j, 1, M}];
20 Horig = H0 + Vx + Vy;
21
22
23 (* =====
24    Compute the Transformation
25    ===== *)
26
27 U = Cos[-lambda[2] * Omegax[t] / (2*Delta)] *
   (ketbra[1,1]+ketbra[2,2]) + I * sigY[2] * Sin[-lambda[2] *
   Omegax[t] / (2*Delta)];
28 U = U + ketbra[0,0];
29 Hpr = U . Horig . ConjugateTranspose[U] + I * D[U,t] .
   ConjugateTranspose[U];
30 (U . Horig . ConjugateTranspose[U])[[1,2]];

```

31 `MatrixForm[Map[Simplify, Hpr[[2,3]], {2}]];`